

HW 1

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1

- (a) Given random variable X where $a \leq X \leq b$, we can prove that $X - E[X]$ is $\frac{(b-a)^2}{4}$ -subgaussian. Let $Y = X - E[X]$ and the log MGF is $\psi(\lambda) = \log E[e^{\lambda Y}]$. Using the identity from the hint, we can bound $\psi''(\lambda)$ as follows:

$$\begin{aligned}\text{Let } M(\lambda) &= E[e^{\lambda Y}] \\ \psi(\lambda) &= \log M(\lambda) \\ \psi'(\lambda) &= \frac{M'(\lambda)}{M(\lambda)} = \frac{E[Y e^{\lambda Y}]}{E[e^{\lambda Y}]} \\ \psi''(\lambda) &= \frac{M''(\lambda)M(\lambda) - (M'(\lambda))^2}{(M(\lambda))^2} = \frac{E[Y^2 e^{\lambda Y}]E[e^{\lambda Y}] - (E[Y e^{\lambda Y}])^2}{(E[e^{\lambda Y}])^2}\end{aligned}$$

Since $Y = X - E[X]$ and subtracting a constant does not change variance:

$$\psi''(\lambda) = \frac{E[Y^2 e^{\lambda Y}]}{E[e^{\lambda Y}]} - \left(\frac{E[Y e^{\lambda Y}]}{E[e^{\lambda Y}]} \right)^2 = \text{Var}(\tilde{X}) \leq \frac{(b-a)^2}{4}$$

where $\tilde{X}$ is a random variable that satisfies $a \leq \tilde{X} \leq b$. Therefore, we have:

$$\psi(\lambda) \leq \int_0^\lambda \int_0^t \frac{(b-a)^2}{4} ds dt = \frac{(b-a)^2 \lambda^2}{8}$$

A mean zero random variable is σ^2 -subgaussian if $\psi(\lambda) \leq \frac{\sigma^2 \lambda^2}{2}$. Thus, we have:

$$\begin{aligned}\frac{\sigma^2 \lambda^2}{2} &= \frac{(b-a)^2 \lambda^2}{8} \\ \sigma^2 &= \frac{(b-a)^2}{4}\end{aligned}$$

Thus, $X - E[X]$ is $\frac{(b-a)^2}{4}$ -subgaussian.

- (b) The constant $\frac{(b-a)^2}{4}$ is the best possible constant. To show this, we use the example of a random variable X that takes on the values a and b with equal probability. Then, we have:

$$E[X] = \frac{a+b}{2}$$

$$Var(X) = E[X^2] - (E[X])^2 = \frac{a^2+b^2}{2} - \left(\frac{a+b}{2}\right)^2 = \frac{(b-a)^2}{4}$$

Let $Y = X - E[X]$. Then, we have:

$$E[Y] = E[X - E[X]] = E[X] - E[X] = 0$$

$$Var(Y) = Var(X) = \frac{(b-a)^2}{4}$$

Now, suppose Y is σ^2 -subgaussian. Then

$$\psi_Y(\lambda) = \log E[e^{\lambda Y}] \leq \frac{\sigma^2 \lambda^2}{2}$$

Since $\psi_Y''(0) = Var(Y)$ and $\left(\frac{\sigma^2 \lambda^2}{2}\right)'' = \sigma^2$, the subgaussian bound near $\lambda = 0$ implies

$$\frac{(b-a)^2}{4} \leq \sigma^2$$

Thus, $Y = X - E[X]$ cannot be σ^2 -subgaussian for any $\sigma^2 < \frac{(b-a)^2}{4}$. Therefore, the constant $\frac{(b-a)^2}{4}$ cannot be improved in general.

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Let X be a random variable where $E[X] = 0$, $Var(X) \leq \sigma^2$, and $|X| \leq b$ almost surely for a finite constant b . For any $\lambda \in (-3/b, 3/b)$, we can show that $\psi(\lambda) \leq \frac{\lambda^2 \sigma^2}{2(1-|\lambda|b/3)}$ by using the Taylor expansion of $e^{\lambda X}$:

$$e^{\lambda X} = 1 + \lambda X + \frac{\lambda^2 X^2}{2!} + \frac{\lambda^3 X^3}{3!} + \dots$$

$$E[e^{\lambda X}] = 1 + \frac{\lambda^2 E[X^2]}{2!} + \frac{\lambda^3 E[X^3]}{3!} + \dots$$

We can define an upper bound $\lambda^n X^n \leq |\lambda|^n |X|^n$, but since $|X| \leq b$, we have $|X|^n \leq b^{n-2} |X|^2$. So:

$$\begin{aligned} E[e^{\lambda X}] &\leq 1 + E[X^2] \left(\frac{\lambda^2}{2!} + \frac{|\lambda|^3 b}{3!} + \frac{|\lambda|^4 b^2}{4!} + \dots \right) \\ E[X] &= 0, \quad E[X^2] = \text{Var}(X) \leq \sigma^2 \therefore \\ E[e^{\lambda X}] &\leq 1 + \sigma^2 \left(\frac{\lambda^2}{2!} + \frac{|\lambda|^3 b}{3!} + \frac{|\lambda|^4 b^2}{4!} + \dots \right) \\ E[e^{\lambda X}] &\leq 1 + \sigma^2 \left(\frac{\lambda^2}{2} \right) \left(1 + \frac{|\lambda| b}{3} + \frac{(|\lambda| b)^2}{12} + \dots \right) \end{aligned}$$

But we can recognize the series as a geometric series for the third term onward since $\frac{|\lambda|^4 b^2}{4!} \leq \frac{\lambda^2 b^2}{2} \left(\frac{|\lambda| b}{3} \right)^2$. Thus, we can write:

$$E[e^{\lambda X}] \leq 1 + \frac{\lambda^2 \sigma^2}{2} \left(1 + \frac{|\lambda| b}{3} + \left(\frac{|\lambda| b}{3} \right)^2 + \dots \right)$$

We know that $|\lambda| < \frac{3}{b}$, so $\frac{|\lambda| b}{3} < 1$ and we can sum the geometric series using $r = \frac{|\lambda| b}{3}$:

$$\begin{aligned} E[e^{\lambda X}] &\leq 1 + \frac{\lambda^2 \sigma^2}{2} \left(\frac{1}{1 - |\lambda| b / 3} \right) \\ E[e^{\lambda X}] &\leq 1 + \frac{\lambda^2 \sigma^2}{2(1 - |\lambda| b / 3)} \\ \log(1 + x) &\leq x \therefore \\ \log E[e^{\lambda X}] = \psi(\lambda) &\leq \frac{\lambda^2 \sigma^2}{2(1 - |\lambda| b / 3)} \end{aligned}$$

3

- (a) Since X is a mean-zero random variable, and $X_+ = \max(X, 0)$, for any $t > 0$:

$$\begin{aligned} P[X \geq t] &= P[X_+ \geq t] \\ P[X_+ \geq t] &= P[X_+^p \geq t^p] \end{aligned}$$

By Markov's inequality, $P[X_+^p \geq t^p] \leq \frac{E[X_+^p]}{t^p}$

$$P[X_+ \geq t] \leq \inf_{p>0} \frac{E[X_+^p]}{t^p}$$

For the second inequality, using $p \rightarrow 0_+$, we get that for each fixed x :

$$\begin{aligned} x_+^p &\rightarrow \begin{cases} 1, & x > 0 \\ 0, & x \leq 0 \end{cases} \therefore \\ X_+^p &\rightarrow \mathbf{1}_{X>0} \therefore \\ \frac{E[X_+^p]}{t^p} &\rightarrow \frac{P[X > 0]}{1} = P[X > 0] \end{aligned}$$

Let $p = \lambda t$ for $\lambda > 0$. Then, since $t > 0$, we have $p > 0$, thus:

$$\begin{aligned} \inf_{p>0} \frac{E[X_+^p]}{t^p} &\leq \frac{E[X_+^{\lambda t}]}{t^{\lambda t}} \\ \frac{x_+^{\lambda t}}{t^{\lambda t}} &= \left(\frac{x}{t}\right)^{\lambda t} = e^{\lambda t \log(x/t)} \\ \log\left(\frac{x}{t}\right) &\leq \frac{x}{t} - 1 \\ \lambda t \log\left(\frac{x}{t}\right) &\leq \lambda(x - t) \\ \left(\frac{x}{t}\right)^{\lambda t} &\leq e^{\lambda(x-t)} \therefore \\ \frac{x_+^{\lambda t}}{t^{\lambda t}} &\leq e^{\lambda(x-t)} \therefore \\ \frac{E[X_+^{\lambda t}]}{t^{\lambda t}} &\leq E[e^{\lambda(X-t)}] = e^{-\lambda t} E[e^{\lambda X}] \therefore \\ \inf_{p>0} \frac{E[X_+^p]}{t^p} &\leq e^{-\lambda t} E[e^{\lambda X}] \quad \forall \lambda > 0 \end{aligned}$$

For $\lambda = 0$, we have from the hint $e^{-\lambda t} E[e^{\lambda X}] = 1 \geq P[X > 0]$. Since the inequality holds over all $\lambda \geq 0$, we can take the infimum over $\lambda \geq 0$:

$$\inf_{p>0} \frac{E[X_+^p]}{t^p} \leq \inf_{\lambda \geq 0} e^{-\lambda t} E[e^{\lambda X}]$$

which gives us the full inequality:

$$P[X \geq t] \leq \inf_{p>0} \frac{E[X_+^p]}{t^p} \leq \inf_{\lambda \geq 0} e^{-\lambda t} E[e^{\lambda X}]$$

- (b) Let $X = \epsilon Y$ where ϵ is a Rademacher random variable independent of Y , where $P[Y \geq y] = y^{-\alpha}$ for $y \geq 1$ and $\alpha > 1$. For $t \geq e$, the exact tail probability $P[X \geq t]$ is:

$$\begin{aligned} P[X \geq t] &= P[\epsilon Y \geq t] = P[\epsilon = 1, Y \geq t] + P[\epsilon = -1, Y \geq t] \\ P[X \geq t] &= \frac{1}{2} P[Y \geq t] + 0 = \frac{1}{2} t^{-\alpha} \end{aligned}$$

For the first infimum, the exact tail probability $P[X \geq t]$ is:

$$\begin{aligned}
X_+ &= \begin{cases} Y, & \epsilon = 1 \\ 0, & \epsilon = -1 \end{cases} \therefore \\
E[X_+^p] &= \frac{1}{2}E[Y^p] + \frac{1}{2}E[0^p] = \frac{1}{2}E[Y^p] \\
E[Y^p] &= \int_1^\infty y^p \alpha y^{-\alpha-1} dy = \frac{\alpha}{\alpha-p} \quad \text{for } 0 < p < \alpha \therefore \\
\inf_{p>0} \frac{E[X_+^p]}{t^p} &= \frac{\alpha}{2(\alpha-p)t^p}
\end{aligned}$$

We need to minimize this:

$$\begin{aligned}
\log\left(\frac{\alpha}{2(\alpha-p)t^p}\right) &= \log\left(\frac{\alpha}{2}\right) - \log(\alpha-p) - p \log(t) \\
\frac{d}{dp} \left(\log\left(\frac{\alpha}{2}\right) - \log(\alpha-p) - p \log(t) \right) &= 0 \\
\frac{1}{\alpha-p} - \log(t) &= 0 \therefore \\
p &= \alpha - \frac{1}{\log(t)} \therefore \\
\inf_{p>0} \frac{E[X_+^p]}{t^p} &= \frac{\alpha}{2(1/\log(t))t^{\alpha-1/\log(t)}} = \frac{\alpha \log(t)}{2} t^{-\alpha+1/\log(t)} = \frac{e\alpha \log(t)}{2et^\alpha}
\end{aligned}$$

Compared to the original tail probability $P[X \geq t] = \frac{1}{2}t^{-\alpha}$, the overestimate factor is:

$$\frac{e\alpha \log(t)}{2et^\alpha} \cdot \frac{2t^\alpha}{1} = e\alpha \log(t)$$

For the second infimum, we have:

$$E[e^{\lambda X}] = \frac{1}{2}E[e^{\lambda Y}] + \frac{1}{2}E[e^{-\lambda Y}]$$

But $E[e^{\lambda Y}] = \int_1^\infty e^{\lambda y} \alpha y^{-\alpha-1} dy$ diverges for any $\lambda > 0$, so we can only consider $\lambda = 0$. Thus, the second infimum is:

$$\inf_{\lambda \geq 0} e^{-\lambda t} E[e^{\lambda X}] = 1$$

Compared to the original tail probability $P[X \geq t] = \frac{1}{2}t^{-\alpha}$, the overestimate factor is:

$$1 \cdot \frac{2t^\alpha}{1} = 2t^\alpha$$

4

(a) If $X = 0$, then for $E[e^{X^2/K^2}] = E[e^0] = 1 \leq 2$, therefore $\|X\|_{\psi_2} = \inf\{K > 0 : E[e^{X^2/K^2}] \leq 2\} = 0$. Suppose $X \neq 0$, then for $K \rightarrow 0^+$, $e^{X^2/K^2} \rightarrow \infty$ and consequently $E[e^{X^2/K^2}] \rightarrow \infty > 2$. Thus, $\|X\|_{\psi_2} > 0$, which contradicts $\|X\|_{\psi_2} = 0$. Therefore, $\|X\|_{\psi_2} = 0$ if and only if $X = 0$.

(b) $\|cX\|_{\psi_2} = |c|\|X\|_{\psi_2}$. If $c = 0$, both sides are zero. If $c \neq 0$, then:

$$\begin{aligned} \|cX\|_{\psi_2} &= \inf\{K > 0 : E[e^{(cX)^2/K^2}] \leq 2\} \\ \|cX\|_{\psi_2} &= \inf\{K > 0 : E[e^{X^2/(K/|c|)^2}] \leq 2\}, \quad K' = K/|c| \\ \|cX\|_{\psi_2} &= |c| \inf\{K' > 0 : E[e^{X^2/(K')^2}] \leq 2\} = |c|\|X\|_{\psi_2} \end{aligned}$$

(c) Let $a > \|X\|_{\psi_2}$ and $b > \|Y\|_{\psi_2}$. Then, we have:

$$\begin{aligned} E[e^{X^2/a^2}] &\leq 2, \quad E[e^{Y^2/b^2}] \leq 2 \\ E[e^{(X+Y)^2/(a+b)^2}] &= E[e^{((a/(a+b))(X/a) + (b/(a+b))(Y/b))^2}] \\ \text{Let } \theta &= \frac{a}{a+b}, \quad 1 - \theta = \frac{b}{a+b} \\ (\theta(X/a) + (1-\theta)(Y/b))^2 &\leq (\theta)(X/a)^2 + (1-\theta)(Y/b)^2 \\ E[e^{(X+Y)^2/(a+b)^2}] &\leq E[e^{(\theta)(X/a)^2 + (1-\theta)(Y/b)^2}] \\ E[e^{(X+Y)^2/(a+b)^2}] &\leq E[e^{(\theta)(X/a)^2}] E[e^{(1-\theta)(Y/b)^2}] \\ E[e^{(X+Y)^2/(a+b)^2}] &\leq E[e^{(\theta)(X/a)^2}] E[e^{(1-\theta)(Y/b)^2}] \\ E[e^{(X+Y)^2/(a+b)^2}] &\leq (E[e^{(X/a)^2}])^\theta (E[e^{(Y/b)^2}])^{1-\theta} \\ E[e^{(X+Y)^2/(a+b)^2}] &\leq (E[e^{(X/a)^2}])^\theta (E[e^{(Y/b)^2}])^{1-\theta} \\ E[e^{(X+Y)^2/(a+b)^2}] &\leq 2^\theta 2^{1-\theta} = 2, \therefore \\ \|X+Y\|_{\psi_2} &\leq a+b = \|X\|_{\psi_2} + \|Y\|_{\psi_2} \end{aligned}$$

5

(a) Let X be a random variable where $E[X] = 0$, $\text{Var}[X] = \sigma^2$, and $X \leq b$ with probability 1. Then, we can show that for all $\lambda \geq 0$:

$$\log E[e^{\lambda X}] \leq \frac{\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2}$$

by starting with $e^{\lambda X} = 1 + \lambda X + (e^{\lambda X} - 1 - \lambda X)$ and bounding the last term. Because $X \leq b$ and $\lambda \geq 0$, $\lambda X \leq \lambda b$. From the hint, we can then

know that:

$$\begin{aligned}
\frac{e^{\lambda X} - 1 - \lambda X}{(\lambda X)^2} &\leq \frac{e^{\lambda b} - 1 - \lambda b}{(\lambda b)^2} \\
e^{\lambda X} - 1 - \lambda X &\leq \frac{X^2(e^{\lambda b} - 1 - \lambda b)}{b^2} \\
e^{\lambda X} &\leq 1 + \lambda X + \frac{X^2(e^{\lambda b} - 1 - \lambda b)}{b^2} \\
E[e^{\lambda X}] &\leq 1 + \lambda E[X] + \frac{E[X^2](e^{\lambda b} - 1 - \lambda b)}{b^2} \\
E[X] &= 0, \quad E[X^2] = \text{Var}[X] = \sigma^2 \therefore \\
E[e^{\lambda X}] &\leq 1 + \frac{\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2} \\
\log(E[e^{\lambda X}]) &\leq \log\left(1 + \frac{\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2}\right) \\
\log\left(1 + \frac{\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2}\right) &\leq \frac{\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2} \therefore \\
\log(E[e^{\lambda X}]) &\leq \frac{\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2}
\end{aligned}$$

- (b) Now, with $X_1, \dots, X_n$ independent random variables equal in law to X , for all $t \geq 0$, we can show that:

$$P\left[\sum_{i=1}^n X_i \geq t\right] \leq \exp\left(-\frac{n\sigma^2}{b^2} h\left(\frac{bt}{n\sigma^2}\right)\right)$$

where $h(x) = (1+x)\log(1+x) - x$ by using the Chernoff bound. From part (a) and because each X_i is independent and equal in law to X , we have:

$$\begin{aligned}
E[e^{\lambda \sum_{i=1}^n X_i}] &= E\left[\prod_{i=1}^n e^{\lambda X_i}\right] = (E[e^{\lambda X}])^n \\
\log E[e^{\lambda \sum_{i=1}^n X_i}] &= n \log E[e^{\lambda X}] \leq \frac{n\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2} \therefore \\
E[e^{\lambda \sum_{i=1}^n X_i}] &\leq \exp\left(\frac{n\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2}\right)
\end{aligned}$$

Now using the Chernoff bound, we have:

$$P\left[\sum_{i=1}^n X_i \geq t\right] \leq \exp\left(-\lambda t + \frac{n\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2}\right)$$

Minimizing the right-hand side over $\lambda \geq 0$, we have:

$$\begin{aligned}\frac{d}{d\lambda} \left(-\lambda t + \frac{n\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2} \right) &= 0 \\ -t + \frac{n\sigma^2(e^{\lambda b} b - b)}{b^2} &= 0 \therefore \\ e^{\lambda b} &= 1 + \frac{bt}{n\sigma^2} \therefore \\ \lambda^* &= \frac{1}{b} \log \left(1 + \frac{bt}{n\sigma^2} \right)\end{aligned}$$

Let $x = \frac{bt}{n\sigma^2}$. Then, we have:

$$\begin{aligned}\lambda^* &= \log(1+x), \quad e^{\lambda^* b} = 1+x, \quad t = \frac{n\sigma^2 x}{b} \\ &\therefore \\ -\lambda^* t &= -\frac{n\sigma^2 x \log(1+x)}{b^2} \\ \frac{n\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2} &= \frac{n\sigma^2((1+x) - 1 - \log(1+x))}{b^2} = \frac{n\sigma^2(x - \log(1+x))}{b^2} \\ &\therefore \\ -\lambda t + \frac{n\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2} &= -\frac{n\sigma^2 x \log(1+x)}{b^2} + \frac{n\sigma^2(x - \log(1+x))}{b^2} \\ -\lambda t + \frac{n\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2} &= -\frac{n\sigma^2}{b^2} (x \log(1+x) + x - \log(1+x)) \\ -\lambda t + \frac{n\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2} &= -\frac{n\sigma^2}{b^2} ((1+x) \log(1+x) - x) \\ -\lambda t + \frac{n\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2} &= -\frac{n\sigma^2}{b^2} h(x)\end{aligned}$$

Then finally, substituting back $x = \frac{bt}{n\sigma^2}$, we have:

$$P \left[\sum_{i=1}^n X_i \geq t \right] \leq \exp \left(-\frac{n\sigma^2}{b^2} h \left(\frac{bt}{n\sigma^2} \right) \right)$$